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Department of Mathematics
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Math 102 Final Exam Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. On the interval $[0, 2\sqrt{3}]$, $\lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} c_i \sqrt{c_i^2 + 4}\, \Delta x_i =$

(a) $20\sqrt{3}$

(b) 20

(c) $\dfrac{56}{3}$

(d) 24

(e) $24\sqrt{3}$

2. The series $\sum_{n=2}^{\infty} \frac{n^2}{n^3+1}$ is

(a) divergent by the Integral Test

(b) convergent by the Comparison Test

(c) divergent by the nth-Term Test

(d) convergent by the Integral Test

(e) convergent by the Limit Comparison Test

3. The area of the surface formed by revolving the graph of $y = \sqrt{25 - x^2}$, $-1 \le x \le 4$ about the x-axis is equal to

(a) 55π

(b) 65π

(c) 60π

(d) 50π

(e) 45π

4. Using the Midpoint Rule with $n = 3$, the approximation of the area of the region bounded by the graph of $f(x) = \sin x$ and the x-axis for $0 \le x \le \pi$ is equal to

(a) $(\sqrt{3}+1)\pi$

(b) $\dfrac{2\pi}{3}$

(c) $\dfrac{3\pi}{5}$

(d) $\dfrac{3\pi}{4}$

(e) $\left(\dfrac{\sqrt{3}+1}{2}\right)\pi$

5. The series $\sum_{n=1}^{\infty} \frac{1}{1+2+3+4+\cdots+n}$

(a) diverges by the integral test

(b) diverges by the nth term test for divergence

(c) converges by the limit comparison test

(d) diverges by the comparison test

(e) converges by the ratio test

6. The sequence $\left\{\dfrac{1+(-1)^n}{n^3}\right\}_{n=1}^{\infty}$

(a) converges to 1

(b) converges to 0

(c) diverges

(d) converges to -1

(e) is not bounded

7. Which of the following statements is true about the two series

$$\text{(I)} \sum_{n=0}^{\infty} \frac{(n!)^3}{(3n)!}, \qquad \text{(II)} \sum_{n=1}^{\infty} \left(\frac{\ln n}{n}\right)^n$$

(a) The series (I) converges but the series (II) diverges

(b) Both series diverge

(c) Both series converge

(d) The series (I) diverges but the series (II) converges

(e) The sequence of partial sums of (II) is not monotonic

8. A power series centered at 0 for $f(x) = \dfrac{2x+1}{x^2-1}$ is

(a) $\sum_{n=0}^{\infty} \left(x^{2n} - 2x^{(2n+1)}\right)$

(b) $-\sum_{n=0}^{\infty} \left(x^{2n} + 2x^{(2n+1)}\right)$

(c) $\sum_{n=0}^{\infty} \left(-2x^{2n} + x^{(2n+1)}\right)$

(d) $\sum_{n=0}^{\infty} \left(x^{2n} + 2x^{(2n+1)}\right)$

(e) $\sum_{n=0}^{\infty} \left(2x^{2n} - x^{(2n+1)}\right)$

9. The geometric series $6 - 3 + \dfrac{3}{2} - \dfrac{3}{4} + \cdots$

(a) is convergent and its sum is $\dfrac{9}{2}$

(b) is convergent and its sum is $\dfrac{7}{2}$

(c) is convergent and its sum is 5

(d) is convergent and its sum is 4

(e) is divergent

10. The series $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt[3]{n}}{n+1}$

(a) is monotonic

(b) is not bounded

(c) converges absolutely

(d) converges conditionally

(e) diverges

11. Using a fourth Maclaurin polynomial of the function $f(x) = e^{4x}$, the approximation of the value of e is equal to

(a) $\dfrac{61}{24}$

(b) $\dfrac{65}{24}$

(c) $\dfrac{63}{24}$

(d) $\dfrac{69}{24}$

(e) $\dfrac{67}{24}$

12. The Maclaurin Series for the function $f(x) = e^{\frac{x^2}{2}}$ is

(a) $\sum_{n=0}^{\infty} \frac{x^{n^2}}{2^n\, n!}$

(b) $\sum_{n=0}^{\infty} \frac{x^{2n}}{2^n\, n!}$

(c) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^n\, n!}$

(d) $\sum_{n=0}^{\infty} \frac{x^{2n}}{2^{n+1}\, n!}$

(e) $\sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$

13. Which of the following statements is true about the two series

$$\text{(I)} \sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right), \qquad \text{(II)} \sum_{n=1}^{\infty} (\sin 2)^n$$

(a) (I) is a convergent series but (II) is a divergent series

(b) The sequence of partial sums of (I) is decreasing

(c) (I) is a divergent series but (II) is a convergent series

(d) Both series are convergent

(e) Both series are divergent

14. The improper integral $\int_0^5 \frac{10}{x}\, dx$

(a) diverges

(b) converges to $5 \ln 5$

(c) converges to $10 \ln 5$

(d) converges to $\ln 5$

(e) converges to $2 \ln 5$

15. $\int_{-\pi/3}^{\pi/3} \left(\sin^3 x \cos x + \sec^2 x \tan^2 x\right) dx =$

(a) $\sqrt{3}$

(b) $\dfrac{\sqrt{3}}{2}$

(c) $2\sqrt{3}$

(d) $\dfrac{2\sqrt{3}}{3}$

(e) 0

16. The sequence $\left\{\dfrac{n^3+1}{n^2+1}\right\}_{n=1}^{\infty}$ is

(a) monotonic but not bounded

(b) monotonic and bounded

(c) neither monotonic nor bounded

(d) bounded but not monotonic

(e) convergent

17. The interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$ is

(a) $[-1, 1]$

(b) $(-1, 1]$

(c) $(-1, 1)$

(d) $[-1, 1)$

(e) $(-\infty, \infty)$

18. $\int \cos^3(x) \sin^4(x)\, dx =$

(a) $\dfrac{\sin^5(x)}{5} - \dfrac{\sin^7(x)}{7} + C$

(b) $-\dfrac{\sin^5(x)}{5} + \dfrac{\sin^7(x)}{7} + C$

(c) $\dfrac{\sin^5(x)}{5} - \dfrac{\sin^3(x)}{3} + C$

(d) $\dfrac{\sin^9(x)}{5} - \dfrac{\sin^8(x)}{7} + C$

(e) $-\dfrac{\sin^5(x)}{5} + \dfrac{\sin^3(x)}{3} + C$

19. The first three terms of the Taylor series generated by the function $f(x) = \dfrac{1}{4-x}$ centered at $c = 2$ are

(a) $\dfrac{1}{2} + \dfrac{1}{4}(x-2) + \dfrac{1}{8}(x-2)^2$

(b) $\dfrac{1}{2} + \dfrac{1}{8}(x-2) + \dfrac{1}{16}(x-2)^2$

(c) $2 + 2(x-4) - 4(x-4)^2$

(d) $\dfrac{1}{2} - \dfrac{1}{4}(x-2) + \dfrac{1}{16}(x-2)^2$

(e) $\dfrac{1}{2} + \dfrac{1}{4}(x-2) - \dfrac{1}{4}(x-2)^2$

20. $\int_{-1}^{6} |5 - 2x|\, dx =$

(a) $\dfrac{45}{2}$

(b) $\dfrac{73}{4}$

(c) $\dfrac{65}{4}$

(d) $\dfrac{49}{2}$

(e) $\dfrac{39}{2}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Limit $= \int_0^{2\sqrt{3}} x\sqrt{x^2+4}\,dx$, $u = x^2+4$: $\frac{1}{3}(64-8) = \frac{56}{3}$.
2	**(a)**	f positive and decreasing; $\int_2^\infty \frac{x^2}{x^3+1}dx = \infty$, so diverges by Integral Test.
3	**(d)**	$y\sqrt{1+y'^2} = 5$, so $S = 2\pi\int_{-1}^{4} 5\,dx = 50\pi$.
4	**(b)**	$\Delta x = \frac{\pi}{3}$, midpoints $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$: $\frac{\pi}{3}(\frac{1}{2}+1+\frac{1}{2}) = \frac{2\pi}{3}$.
5	**(c)**	$a_n = \frac{2}{n(n+1)}$; limit comparison with $\sum \frac{1}{n^2}$ (ratio $\to 2$): converges.
6	**(b)**	$0 \le a_n \le \frac{2}{n^3} \to 0$, so by squeeze the limit is 0.
7	**(c)**	(I) ratio test: $L = \frac{1}{27}$; (II) root test: $L = 0$. Both converge.
8	**(b)**	$f = -(2x+1)\frac{1}{1-x^2} = -\sum(x^{2n} + 2x^{2n+1})$.
9	**(d)**	$a = 6$, $r = -\frac{1}{2}$: sum $= \frac{6}{1+1/2} = 4$.
10	**(d)**	AST holds ($b_n \downarrow 0$); $\sum \lvert a_n\rvert \sim \sum n^{-2/3}$ diverges: conditionally convergent.
11	**(b)**	$4x = 1$: $e \approx 1+1+\frac{1}{2}+\frac{1}{6}+\frac{1}{24} = \frac{65}{24}$.
12	**(b)**	e^u with $u = \frac{x^2}{2}$: $\sum \frac{x^{2n}}{2^n n!}$.
13	**(c)**	(I) telescopes to $\ln(N+1) \to \infty$; (II) is geometric with $\lvert\sin 2\rvert < 1$.
14	**(a)**	$\lim_{t\to 0^+} 10(\ln 5 - \ln t) = \infty$, so it diverges.
15	**(c)**	First term is odd (integral 0); $2\left[\frac{\tan^3 x}{3}\right]_0^{\pi/3} = 2\sqrt{3}$.
16	**(a)**	$a_n \sim n \to \infty$ and a_n is increasing: monotonic, not bounded.
17	**(a)**	$R = 1$; both endpoints give $\pm\sum \frac{(-1)^n}{2n+1}$ (AST): $[-1, 1]$.
18	**(a)**	$u = \sin x$: $\int (u^4 - u^6)\,du = \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C$.
19	**(a)**	$f(2) = \frac{1}{2}$, $f'(2) = \frac{1}{4}$, $f''(2)/2! = \frac{1}{8}$.
20	**(d)**	Split at $x = \frac{5}{2}$: $\frac{49}{4} + \frac{49}{4} = \frac{49}{2}$.